

# ELI — Code Mode Execution Layer

ELI v2.3.7

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## Blueprint — Code-Mode Execution Layer for ELI (gap analysis)

*Status: draft. Date: 2026-06-11. REVISED after auditing the codebase: an earlier version of this doc proposed building machinery ELI already has. This version is a GAP ANALYSIS — what exists, and the small genuine delta — not a greenfield design. Original ELI work throughout; the only outside idea is the general principle of deterministic gating, which ELI already implements its own way.*

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### BUILT (2026-06-11)

Core implemented, gated, and tested (17 tests, tests/test\_code\_mode.py): - **Restricted exec (the one new piece):** eli/coding/restricted\_exec.py — Full-Control-gated (is\_full\_control(), the GUI toggle), AST-whitelist (10 escape classes rejected: import/ eval/open/getattr/dunder-attr/\_\_import\_\_/def/lambda/with/exec), restricted namespace. - **Facade (sugar):** eli/tools/api.py — api.call("ACTION", \*\*args) + helpers, proxying to execute(); api.actions() for discovery. - **Runner:** eli/coding/code\_mode.py — generate→gate+validate+run→retry loop (injectable generate, testable without a model). - **Lane redirect:** CODE\_SOLVE with args.code\_mode=True runs in-process against eli.api (fail-closed: refuses unless Full Control is ON). - **#4 retrieval:** confirmed ABSENT (no top-K capability retriever); v1 passes a bounded api.actions() list — a semantic retriever is a future refinement, not required.

Remaining (deferred, behaviour-policy): the router auto-selecting code-mode for multi-step requests under Full Control — needs live validation, so it's flag-reachable for now.

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### 0. Correction up front

“Code-mode” (the model writes a small program against a curated API instead of emitting MCP-style tool-calls) is the token-frugal, composable, offline-friendly direction the field is moving to — and the right fit

for ELI's 8 GB / small-context reality. **But ELI already has ~85% of the substrate.** This doc exists to name the ~15% that's actually new, so we build only that.

## 1. What ELI ALREADY HAS (audited, with files)

| Capability the blueprint needs                                                           | Already in ELI                                                                    | File                                                            |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Code-agent loop: plan → implement → verify → run → retry                                 | <b>CodeAgent.solve()</b> — “orchestrator that composes ELI's coding capabilities” | eli/coding/agent.py                                             |
| Planner / implementer separation                                                         | plan_task(), implement()                                                          | eli/coding/planner.py                                           |
| Candidate verification + <b>auto-synthesised tests</b>                                   | verify_candidate(), synthesize_tests()                                            | eli/coding/verification.py                                      |
| Tree search over candidates                                                              | tree_search()                                                                     | eli/coding/search.py                                            |
| Learn from failed attempts                                                               | BugMemory                                                                         | eli/coding/bug_memory.py                                        |
| DAG decomposition of a task                                                              | solve_dag()                                                                       | eli/coding/plan_graph.py                                        |
| <b>Bounded sandbox execution</b> (CPU limit, scrubbed env, kill-switch)                  | run_code()                                                                        | eli/coding/sandbox.py                                           |
| <b>Deterministic risk gate</b> (action classes → approved/blocked/pending-confirm)       | approval_engine.evaluate_record()                                                 | eli/runtime/approval_engine.py                                  |
| <b>Fail-closed command allowlist</b> + destructive-pattern block (rm -rf /, mkfs, dd...) | security + _BLOCKED_PATTERNS                                                      | eli/runtime/security.py, executor_enhanced.py                   |
| Authority / autonomy posture                                                             | authority_gate, operator_policy (proposal_only/supervised/...)                    | eli/runtime/authority_gate.py, eli/execution/operator_policy.py |
| AST validate-and-retry of generated code                                                 | code_examiner (used live 2026-06-11)                                              | eli/runtime/code_examiner.py                                    |
| Capability <b>registry</b> (register/get/list)                                           | capability_registry                                                               | eli/tools/registry/capability_registry.py                       |
| Capability <b>doc/digest generator</b> from the manifest                                 | generate_capabilities_doc()                                                       | eli/tools/registry/capabilities_doc.py                          |
| Offline enforcement / crisis guard                                                       | netguard, crisis_guard                                                            | eli/core/                                                       |
| Reasoning-native planning (model thinks, then emits)                                     | working post-e7ab0b9                                                              | eli/cognition/gguf_inference.py                                 |

**Conclusion:** ELI is not missing “a code agent,” “a sandbox,” or “a safety gate.” It has all three, plus verification, search, bug-memory, a risk engine, a command allowlist, and a capability registry. The CodeAgent already does CodeAct-style plan→code→verify→sandbox→retry.

## 2. The genuine DELTA (the only new work)

The existing CodeAgent solves **coding tasks** — produce a function/script as a *deliverable*. Code-mode-as-assistant-execution needs it pointed at a different target: the model writes code that **calls ELI's own capabilities** to fulfil a *conversational request*, and the reply is synthesised from the execution result. Each candidate delta was audited IN FULL (2026-06-11); statuses below are grounded with file:line evidence.

1. **eli.api facade — MOSTLY EXISTS (only typed sugar is new).** The callable dispatch already exists: execute(action, args) at executor\_enhanced.py:10653 runs any of the 216 actions by name.

Generated code can already call `execute("SUMMARIZE_FILE", {...})`. The *only* delta is a typed, discoverable wrapper (`eli.api.files.summarize(path)`) generated from the registry — convenience + discoverability over an existing surface, not new execution machinery.

2. **In-process AST-whitelist + restricted exec — GENUINELY NEW (the one real piece).** `sandbox.run_code()` (`coding/sandbox.py:106`) is **process-isolated** (subprocess, scrubbed env, CPU limit) and runs *standalone* code — it deliberately cannot reach ELI's live state (the loaded model, memory stores). Code-mode must run **in-process** to call `live eli.api/execute()`, so it **cannot reuse that sandbox**. So an in-process restricted-exec is required: an AST pass allowing only `eli.api.*/execute(...)` + safe builtins, run in a restricted-globals namespace. It **composes** existing parts — the `code_examiner` AST walker + `approval_engine` action-classes for the GREEN/AMBER/RED decision — but the in-process restricted-exec itself is new. This is THE work.
3. **Routing lane to the code agent — ALREADY EXISTS as CODE\_SOLVE.** Router lane (`router_enhanced.py:2254, _CODE_SOLVE_RE`) + executor handler (`executor_enhanced.py:8702`) → `eli.coding.solve` (plan→search→verify→repair/quick/thorough, DAG, background). Reuse it. The only variant is targeting `eli.api` for assistant-actions rather than producing a standalone script, + grounded synthesis of the result (reuses the 540f514 no-confabulation work). Do NOT build a new lane.
4. **Per-query capability retrieval — UNCONFIRMED; parts exist, no retriever found.** There is a manifest (`capability_inventory.generated.json`), a digest generator (`capabilities_doc.generate_capabilities_capability_sync`), and a FAISS store — but a top-K *capability retriever* was NOT located in the audit. Verify before building; if absent, it's a small assembly from the manifest + existing vector store, not new infra.

**Net after full audit:** #3 exists, #1 is sugar over an existing dispatch, #4 is unconfirmed (parts present), and only **#2 (the in-process restricted-exec) is genuinely new** — and it composes existing AST + gate machinery. The real build is: one in-process AST-whitelist + the typed facade, then point the existing `CODE_SOLVE` path at `eli.api`.

### 3. Phased plan (composing what exists)

- **Phase 0 — the one new piece: in-process restricted-exec (~3-5 days).** AST whitelist (reuse `code_examiner`'s walker) allowing only `eli.api.*/execute(...)` + safe builtins; run in a restricted-globals namespace IN-PROCESS (not `sandbox.run_code()`, which is process-isolated and can't see live state); classify each call via `approval_engine` action classes for the GREEN/AMBER/RED decision. This is the genuinely new safety surface.
- **Phase 1 — facade + point CODE\_SOLVE at it (~2-3 days).** Generate the typed `eli.api` wrapper from the registry over `execute()`. Add a `CODE_SOLVE` *variant* (or arg) that targets `eli.api` for assistant-actions and feeds the result into grounded synthesis — reusing the existing lane, agent, quick/thorough modes, and background handling. No new routing lane, no new agent.
- **Phase 2 — retrieval (if needed) + harden (~few days).** Add top-K capability retrieval ONLY if the audit confirms it's absent; add code-mode cases to the eval harness; add the live integration test the suite is missing.

Total honest estimate: **~1-1.5 weeks**, and the bulk of it is the single new piece (the in-process restricted-exec). The agent loop, sandbox, risk gate, verification, bug-memory, registry, AND the `CODE_SOLVE` routing lane are already yours.

### 4. Verified in full (2026-06-11)

Audited against the project directory, not assumed: - **Callable dispatch surface?** YES — `execute(action, args)` at `executor_enhanced.py:10653`. Generated code can already run any of the 216 actions by name. - **Routing lane to the code agent?** YES — `CODE_SOLVE` (`router_enhanced.py:2254` + `executor_enhanced.py:8702` → `eli.coding.solve`). Already plan→search→verify→repair, quick/thorough, DAG, background. - **Sandbox: namespace-restricted or process-only?** PROCESS-only (`coding/sandbox.py:106`: subprocess + scrubbed env + CPU limit), for *standalone* code. **Cannot host live-state code-mode** → the in-process AST-whitelist (§2.2) is genuinely required. - **Risk gate to compose with?** YES — `approval_engine.evaluate_record()` (action classes → approved/blocked/pending-confirm) + the

command allowlist. Map `eli.api` calls onto it. - **AST walker to reuse?** YES — `code_examiner` (used live this session). - **Existing `eli.api`-style typed facade under `eli/tools//eli/plugins/`?** NOT found — the registry stores metadata, not a callable typed surface. This (small) piece is new. - **Top-K capability retriever?** NOT found — manifest + digest generator + FAISS exist, but no retriever. Verify once more at build time; if truly absent, assemble from existing parts.

## 5. What stays uniquely ELI

Voice, vision, gaze, hard-offline, emergent persona, and the entire coding-agent + gate stack you already built — untouched. This is wiring ELI's own parts into a new lane, not importing anyone's design.